Question ID: 628300a9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question

A science teacher is preparing the 5 stations of a science laboratory. Each station will have either Experiment A materials or Experiment B materials, but not both. Experiment A requires 6 teaspoons of salt, and Experiment B requires 4 teaspoons of salt. If x is the number of stations that will be set up for Experiment A and the remaining stations will be set up for Experiment B, which of the following expressions represents the total number of teaspoons of salt required?

Answer

A. $5x$

B. $10x$

C. $2x + 20$

D. $10x + 20$

Correct Answer: C

Rationale

Choice C is correct. It is given that x represents the number of stations that will be set up for Experiment A and that there will be 5 stations total, so it follows that 5 – x is the number of stations that will be set up for Experiment B. It is also given that Experiment A requires 6 teaspoons of salt and that Experiment B requires 4 teaspoons of salt, so the total number of teaspoons of salt required is 6x + 4(5 – x), which simplifies to 2x + 20.

Choices A, B, and D are incorrect and may be the result of not understanding the description of the context.

Question ID: f5ff91b2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question

If $\frac{x-5}{7} = \frac{x-5}{9}$, the value of $x - 5$ is between which of the following pairs of values?

Answer

A. -9 and -7

B. -3 and 3

C. 4.5 and 5.5

D. 6.75 and 9.25

Correct Answer: B

Rationale

Choice B is correct. Multiplying both sides of the given equation by $(7)(9)$, or 63, yields $(63)\left(\frac{x-5}{7}\right) = (63)\left(\frac{x-5}{9}\right)$, or $9(x-5) = 7(x-5)$. Subtracting $7(x-5)$ from both sides of this equation yields $2(x-5) = 0$. Dividing both sides of this equation by 2 yields $x - 5 = 0$. Therefore, if $\frac{x-5}{7} = \frac{x-5}{9}$, then the value of $x - 5$ is 0. It follows that of the given choices, the value of $x - 5$ is between -3 and 3.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 429fb7c0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question

What value of t is the solution to the equation $0.8t - 0.46 = 8(t - 0.001) + 1.9$?

Correct Answer: -.3266, -.3267, -49/150

Rationale

The correct answer is $-.3267$. Applying the distributive property to the right-hand side of the given equation yields $0.8t - 0.46 = 8t - 0.008 + 1.9$, or $0.8t - 0.46 = 8t + 1.892$. Subtracting $0.8t$ from both sides of this equation yields $-0.46 = 7.2t + 1.892$. Subtracting 1.892 from both sides of this equation yields $-2.352 = 7.2t$. Dividing both sides of this equation by 7.2 yields $\frac{-2.352}{7.2} = t$. Therefore, the value of t is approximately -0.32667. Note that -.3267, -.3266, -0.326, and -0.327 are examples of ways to enter a correct answer.

Question ID: 3f8a701b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question

The equation $9x+5=a(x+b)$, where a and b are constants, has no solutions. Which of the following must be true?

I. $a=9$

II. $b=5$

III. $b \neq \frac{5}{9}$

Answer

A. None

B. I only

C. I and II only

D. I and III only

Correct Answer: D

Rationale

Choice D is correct. For a linear equation in a form $ax+b=cx+d$ to have no solutions, the x-terms must have equal coefficients and the remaining terms must not be equal. Expanding the right-hand side of the given equation yields $9x+5=ax+ab$. Inspecting the x-terms, 9 must equal a, so statement I must be true. Inspecting the remaining terms, 5 can't equal $9b$. Dividing both of these quantities by 9 yields that b can't equal $\frac{5}{9}$. Therefore, statement III must be true. Since b can have any value other than $\frac{5}{9}$, statement II may or may not be true.

Choice A is incorrect. For the given equation to have no solution, both $a=9$ and $b \neq \frac{5}{9}$ must be true. Choice B is incorrect because it must also be true that $b \neq \frac{5}{9}$. Choice C is incorrect because when $a=9$, there are many values of b that lead to an equation having no solution. That is, b might be 5, but b isn't required to be 5.

Question ID: aee9fd2d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question

If $\frac{x+6}{3} = \frac{x+6}{13}$, the value of $x + 6$ is between which of the following pairs of values?

Answer

A. -7 and -3

B. -2 and 2

C. 2 and 7

D. 8 and 13

Correct Answer: B

Rationale

Choice B is correct. Multiplying both sides of the given equation by $(3)(13)$, or 39, yields $(39)\left(\frac{x+6}{3}\right) = (39)\left(\frac{x+6}{13}\right)$, or $13(x + 6) = 3(x + 6)$. Subtracting $3(x + 6)$ from both sides of this equation yields $10(x + 6) = 0$. Dividing both sides of this equation by 10 yields $x + 6 = 0$. Therefore, if $\frac{x+6}{3} = \frac{x+6}{13}$, then the value of $x + 6$ is 0. It follows that of the given choices, the value of $x + 6$ is between -2 and 2.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 771bd0ca

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question

$$5(t+3)-7(t+3)=38$$

What value of t is the solution to the given equation?

Correct Answer: -22

Rationale

The correct answer is -22. The given equation can be rewritten as $-2(t+3)=38$. Dividing both sides of this equation by -2 yields $t+3=-19$. Subtracting 3 from both sides of this equation yields $t=-22$. Therefore, -22 is the value of t that is the solution to the given equation.

Question ID: e6cb2402

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question

$$3(kx+13)=\frac{48}{17}x+36$$

In the given equation, k is a constant. The equation has no solution. What is the value of k?

Correct Answer: .9411, .9412, 16/17

Rationale

The correct answer is $\frac{16}{17}$. It's given that the equation $3(kx+13)=\frac{48}{17}x+36$ has no solution. A linear equation in the form $ax+b=cx+d$, where a, b, c, and d are constants, has no solution only when the coefficients of x on each side of the equation are equal and the constant terms aren't equal. Dividing both sides of the given equation by 3 yields $kx+13=\frac{48}{51}x+\frac{36}{3}$, or $kx+13=\frac{16}{17}x+12$. Since the coefficients of x on each side of the equation must be equal, it follows that the value of k is $\frac{16}{17}$. Note that 16/17, .9411, .9412, and 0.941 are examples of ways to enter a correct answer.

Question ID: 25e1cfed

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question

How many solutions does the equation $10(15x - 9) = -15(6 - 10x)$ have?

Answer

A. Exactly one

B. Exactly two

C. Infinitely many

D. Zero

Correct Answer: C

Rationale

Choice C is correct. Applying the distributive property to each side of the given equation yields $150x - 90 = -90 + 150x$. Applying the commutative property of addition to the right-hand side of this equation yields $150x - 90 = 150x - 90$. Since the two sides of the equation are equivalent, this equation is true for any value of x. Therefore, the given equation has infinitely many solutions.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: b7e6394d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question

Alan drives an average of 100 miles each week. His car can travel an average of 25 miles per gallon of gasoline. Alan would like to reduce his weekly expenditure on gasoline by $5. Assuming gasoline costs $4 per gallon, which equation can Alan use to determine how many fewer average miles, m, he should drive each week?

Answer

A. $\frac{25}{4}m = 95$

B. $\frac{25}{4}m = 5$

C. $\frac{4}{25}m = 95$

D. $\frac{4}{25}m = 5$

Correct Answer: D

Rationale

Choice D is correct. Since gasoline costs $4 per gallon, and since Alan's car travels an average of 25 miles per gallon, the expression $\frac{4}{25}$ gives the cost, in dollars per mile, to drive the car. Multiplying $\frac{4}{25}$ by m gives the cost for Alan to drive m miles in his car. Alan wants to reduce his weekly spending by $5, so setting $\frac{4}{25}$m equal to 5 gives the number of miles, m, by which he must reduce his driving.

Choices A, B, and C are incorrect. Choices A and B transpose the numerator and the denominator in the fraction. The fraction $\frac{25}{4}$ would result in the unit miles per dollar, but the question requires a unit of dollars per mile. Choices A and C set the expression equal to 95 instead of 5, a mistake that may result from a misconception that Alan wants to reduce his driving by 5 miles each week; instead, the question says he wants to reduce his weekly expenditure by $5.

Question ID: f14484a5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question

A manufacturing plant makes 10-inch, 9-inch, and 7-inch frying pans. During a certain day, the number of 10-inch frying pans that the manufacturing plant makes is 4 times the number n of 9-inch frying pans it makes, and the number of 7-inch frying pans it makes is 10. During this day, the manufacturing plant makes 100 frying pans total. Which equation represents this situation?

Answer

A. $10(4n) + 9n + 7(10) = 100$

B. $10n + 9n + 7n = 100$

C. $4n + 10 = 100$

D. $5n + 10 = 100$

Correct Answer: D

Rationale

Choice D is correct. It's given that during a certain day, the number of 9-inch frying pans the manufacturing plant makes is n and the number of 7-inch frying pans it makes is 10. It's also given that during this day the number of 10-inch frying pans that the manufacturing plant makes is 4 times the number of 9-inch frying pans, or $4n$. Therefore, the total number of 7-inch, 9-inch, and 10-inch frying pans the manufacturing plant makes is $n + 10 + 4n$, or $5n + 10$. It's given that during this day the manufacturing plant makes 100 frying pans total. Thus, the equation $5n + 10 = 100$ represents this situation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: 2937ef4f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question

Hector used a tool called an auger to remove corn from a storage bin at a constant rate. The bin contained 24,000 bushels of corn when Hector began to use the auger. After 5 hours of using the auger, 19,350 bushels of corn remained in the bin. If the auger continues to remove corn at this rate, what is the total number of hours Hector will have been using the auger when 12,840 bushels of corn remain in the bin?

Answer

A. 3

B. 7

C. 8

D. 12

Correct Answer: D

Rationale

Choice D is correct. After using the auger for 5 hours, Hector had removed 24,000 – 19,350 = 4,650 bushels of corn from the storage bin. During the 5-hour period, the auger removed corn from the bin at a constant rate of $\frac{4,650}{5} = 930$ bushels per hour. Assuming the auger continues to remove corn at this rate, after x hours it will have removed 930x bushels of corn. Because the bin contained 24,000 bushels of corn when Hector started using the auger, the equation 24,000 – 930x = 12,840 can be used to find the number of hours, x, Hector will have been using the auger when 12,840 bushels of corn remain in the bin. Subtracting 12,840 from both sides of this equation and adding 930x to both sides of the equation yields 11,160 = 930x. Dividing both sides of this equation by 930 yields x = 12. Therefore, Hector will have been using the auger for 12 hours when 12,840 bushels of corn remain in the storage bin.

Choice A is incorrect. Three hours after Hector began using the auger, 24,000 – 3(930) = 21,210 bushels of corn remained, not 12,840. Choice B is incorrect. Seven hours after Hector began using the auger, 24,000 – 7(930) = 17,490 bushels of corn will remain, not 12,840. Choice C is incorrect. Eight hours after Hector began using the auger, 24,000 – 8(930) = 16,560 bushels of corn will remain, not 12,840.

Question ID: ac472881

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question

$$\frac{12x+28}{4} - \frac{s}{13} = r(x-8)$$

In the given equation, s and r are constants, and $s > 0$. If the equation has infinitely many solutions, what is the value of s?

Correct Answer: 403

Rationale

The correct answer is 403. For a linear equation in one variable to have infinitely many solutions, the coefficients of the variable must be equal on both sides of the equation and the constant terms must also be equal on both sides of the equation. The given equation can be rewritten as $\frac{4(3x+7)}{4} - \frac{s}{13} = r(x-8)$, or $3x + 7 - \frac{s}{13} = r(x-8)$. Applying the distributive property to the right-hand side of this equation yields $3x + 7 - \frac{s}{13} = rx - 8r$. For this equation to have infinitely many solutions, the coefficients of x must be equal, so it follows that $3 = r$. Additionally, the constant terms must be equal, which means $7 - \frac{s}{13} = -8r$. Substituting 3 for r in this equation yields $7 - \frac{s}{13} = -8(3)$, or $7 - \frac{s}{13} = -24$. Adding $\frac{s}{13}$ to both sides of this equation yields $7 = -24 + \frac{s}{13}$. Adding 24 to both sides of this equation yields $31 = \frac{s}{13}$. Multiplying both sides of this equation by 13 yields $403 = s$. Therefore, if the equation has infinitely many solutions, the value of s is 403.